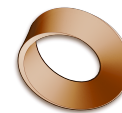


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.11 – Damping and Engineering Solutions

Unit 1: Oscillations & Waves

Learning intentions

- a) I can explain how **damping** removes energy from an oscillating system to reduce amplitude.
- b) I can distinguish between light damping, heavy damping, and critical damping based on system response.
- c) I can describe real-world engineering solutions used to control resonance, such as tuned mass dampers and base isolation.
- d) I can evaluate the trade-offs between the cost of anti-resonance designs and the risks of structural failure.

Theory

When a system oscillates, it possesses mechanical energy that keeps it moving. **Damping** is the process by which this energy is gradually removed from the system, usually converted into heat through friction or air resistance. As energy decreases, the **amplitude** (the maximum displacement from the equilibrium position) of the oscillation reduces over time. Without damping, an ideal system would oscillate forever; in reality, all systems experience some form of damping.

Engineers classify damping into three main types based on how quickly the oscillations stop. **Light damping** occurs when energy is removed slowly, allowing the system to oscillate for many cycles with a gradually decreasing amplitude. This is desirable in pendulum clocks or musical instruments where sustained motion is needed. **Heavy damping** removes energy rapidly, causing the system to return to equilibrium without completing even one full oscillation, or stopping it very quickly after a few cycles. This is useful in car shock absorbers to prevent bouncing. A special case is **critical damping**, where the system returns to equilibrium as quickly as possible without oscillating at all.

Resonance happens when a driving force matches a system's natural frequency, causing large amplitude vibrations that can lead to structural failure. To design against this, engineers use various strategies. One common solution is the **tuned mass damper**, a large counterweight installed in tall buildings like Taipei 101. This mass oscillates out of phase with the building's movement, absorbing energy and reducing sway during high winds or earthquakes. Another method is **base isolation**, where flexible bearings are placed between a building's foundation and the ground to decouple the structure from seismic waves.

In transportation, **springs and dampers** work together in suspension systems. The spring stores energy to absorb bumps, while the damper (shock absorber) provides damping to stop the spring from bouncing indefinitely. Similarly, footbridges may suffer from resonance if pedestrians walk in rhythm with the bridge's natural frequency. Engineers often retrofit these bridges with viscous dampers or adjust their stiffness to change the natural frequency away from the walking pace.

Designing against resonance involves weighing costs against risks. High-performance materials and complex systems like tuned mass dampers are expensive to install and maintain. However, the cost of ignoring resonance can be catastrophic, as seen in historical bridge collapses. Engineers must calculate whether the financial investment in damping solutions is justified by the potential danger to life and property.

Complete the sentence: Engineers use a _____ in tall buildings like Taipei 101 to absorb energy from wind-induced sway and prevent excessive movement.

Activity

Investigating Damping with a Pendulum

You will investigate how different amounts of damping affect the decay of oscillations in a simple pendulum.

What you need: - A clamp stand, string, and a heavy bob (e.g., a metal washer or nut). - A ruler to measure amplitude. - Optional: A piece of card attached to the bob to increase air resistance (for light damping) or a shallow tray of water to place the bob in (for heavy damping).

Steps: 1. Set up the pendulum with a fixed length (e.g., 50 cm). Measure and record this length. 2. Displace the bob to an initial angle of 10 degrees from the vertical. This is your starting **amplitude**. 3. Release the bob and count the number of complete oscillations it takes for the amplitude to halve (i.e., drop to 5 degrees). Record this in the table below as the number of oscillations needed to halve the amplitude. 4. Repeat step 2 and 3 two more times to get an average. 5. Modify the system to increase damping (e.g., attach a large piece of card to the bob to catch more air). 6. Repeat steps 2-4 for this heavily damped system. 7. Compare the number of oscillations required to halve the amplitude in both cases.

Results Table:

Damping Type	Trial 1 (oscillations)	Trial 2 (oscillations)	Trial 3 (oscillations)
Light Damping (no card)	_____	_____	_____
Heavy Damping (with card)	_____	_____	_____

Questions

1. Define **damping** in terms of energy and amplitude.

2. Explain why a car suspension system needs both springs and dampers, rather than just springs.

3. Calculate the period of a pendulum that completes 10 oscillations in 20 seconds. Show your working using $T = t/n$.

4. A bridge collapses because pedestrians walked in step with its natural frequency. Explain how changing the stiffness of the bridge supports would help prevent this in the future.

5. Evaluate one advantage and one disadvantage of installing a tuned mass damper in a skyscraper.

Extension

Research the collapse of the Tacoma Narrows Bridge in 1940. Identify whether the failure was caused by forced resonance or aeroelastic flutter, and explain how modern engineering principles (such as aerodynamic shaping of decks) now prevent this specific type of failure in long-span bridges.
